Advanced statistics

22-01-26

There are two types of data / probability distributions

1) discrete distribution
2) continuous distribution

Discrete Distribution

- these prob. distributions are linked to discrete data type (as well as categorical data type)

- the data which follows discrete distribution arises from process where outcomes are <u>counted</u> rather than measured

examples:

- Number of Heads when a coin is tossed 10 times
  possible values - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
  ↓
  discrete data

- Number of qualified applicants for job
  possible values: 0, 1, 2, 3, 4, 5. - - -

Some important discrete distributions-

1) Bernouli distribution

2) Binomial distribution

$$P(n=6, k=3) = {}^6C_3 \; p^3 (1-p)^{6-3}$$

3) Poisson distribution
- represents the count of events in a given fixed time interval

eg. In a shop, on an average 5 customers come every 30 mins. whats the probability that 10 customers will come in next 45 mins?

→ average rate = 5 customers per 30 mins

λ = 7.5 customers per 45 mins

30 → 45
5 → 7.5

K = number of events = 10 customers

$$P = \frac{\lambda^K e^{-\lambda}}{K!}$$ where e = 2.71 eccentricity constant

$$P(\lambda = 7.5, K = 10) = \frac{(7.5)^{10} e^{-7.5}}{10!}$$

4) Multinomial distribution

- suitable for random experiments with more than two possible outcomes.

eg.
rolling a dice 5 times. whats prob. that '2' will come exactly 3 times out of 5, '3' and '6' will come once?

→ 6 possible outcomes - 1, 2, 3, 4, 5, 6

prob of every outcome = 1/6

2 2 2 3 6, 3 6 2 2 2, 2 2 3 6 2, 6 3 2 2 2, 2 6 3 2 2 . . . .

$$P('2'=3, '3'=1, '6'=1, n=5) = \frac{5!}{0!\,3!\,1!\,0!\,0!\,1!}\left(\frac{1}{6}\right)^0\left(\frac{1}{6}\right)^3\left(\frac{1}{6}\right)^1\left(\frac{1}{6}\right)^0\left(\frac{1}{6}\right)^0\left(\frac{1}{6}\right)^1$$

'1' = 0, '4' = 0, '5' = 0

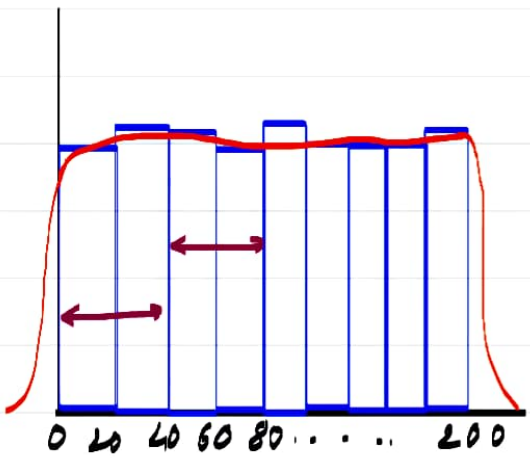
Continuous Distribution

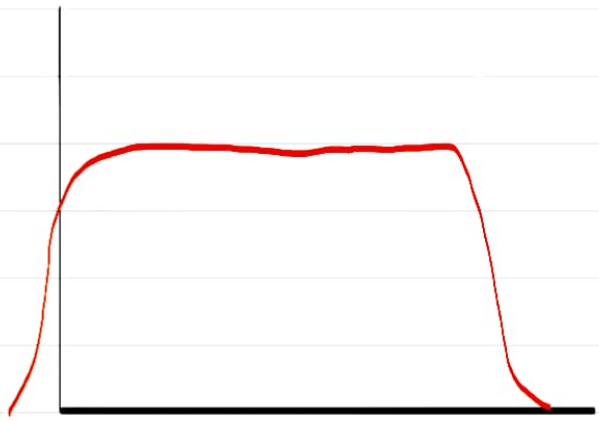
- used when the outcome of process or random experiment is continuous numerical variable
- when dataset is continuous variable

eg. uniform, normal/guassian, exponential, log-normal, pareto, etc

Uniform distribution

eg. we have data of salary of employees in a company.
This data is continuous





- when KDE plot of continuous data is almost flat curve on top then that data is following a uniform distribution

- In such cases, all equal width bins have nearly equal proportion of data

eg. no. of employees with salary 15-45 will be nearly same as
no. of employees with salary 60-90

Probability function of uniform dist

bernouli dist $f(x) = p^{x}(1-p)^{1-x}$
binomial dist $f(n,k) = {}^{n}C_{k}\, p^{k}(1-p)^{n-k}$

} probability mass function PMF

Probability function associated to discrete distribution is called Probability Mass Function PMF

Probability function associated to continuous distribution is called Probability Density Function PDF

PDF for uniform dist:

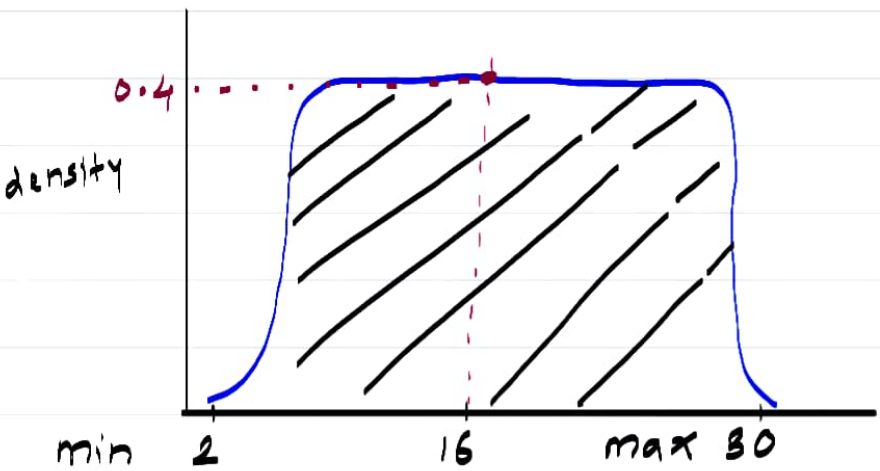
min value in data = a
max value in data = b

$$f(x) = \frac{1}{b-a} \quad \text{if} \quad a \le x \le b$$

$$0 \quad \text{otherwise}$$

eg. data = [12.5, 10.5, 11, 8, 9, ...]

↓ data distribution



area under the KDE curve is 1

↓

integration can compute this

0.4 is not the probability of $x = 16$

0.4 is density value of $x = 16$

We still can compute probability for continuous dist using area under the curve